Unit 4, Lesson 2: Generating Equivalent Numerical Expressions

Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

Learning Target

- ☐ **I can generate equivalent numerical expressions for an expression with a rational base and integer exponent by applying the law of exponents.**

4.2.1 Warm-Up: Math Argument

1. Without performing any calculations, circle the expression you think has the greater value.

60^{63} 63^{60}

2. Justify why you selected that expression.

4.2.2 Guided Instruction: Rational Number Bases

1. Consider the table shown.

Exponential Expression	Expanded Form	Equivalent Expression	
$\left(\frac{2}{5}\right)^3$	$\frac{2}{5}\cdot\frac{2}{5}\cdot\frac{2}{5}$	$\frac{2^3}{5^3}$	$\div\frac{2}{5}$
$\left(\frac{2}{5}\right)^2$	$\frac{2}{5}\cdot\frac{2}{5}$	$\frac{2^2}{5^2}$	$\div\frac{2}{5}$
$\left(\frac{2}{5}\right)^1$	$\frac{2}{5}$	$\frac{2^1}{5^1}$	$\div\frac{2}{5}$
$\left(\frac{2}{5}\right)^0$	1	$\frac{2^0}{5^0}$	

Complete the statement to describe the pattern in the table.

As the value of the exponent
- o increases
- o decreases

by 1 in the column on the left, the equivalent expression in the column on the right can be found by
- o multiplying
- o dividing

the previous value by the
- o base.
- o exponent.

Continuing the pattern, the next row in the table would begin with the expression $\left(\frac{2}{5}\right)^{-1}$.

To continue the pattern, find the value of $1 \div \frac{2}{5}$. Since dividing by a fraction is equivalent to multiplying by the fraction's reciprocal, the result is as follows.

$$1 \div \frac{2}{5} = 1 \cdot \frac{5}{2} = \frac{5}{2}$$

Since $\left(\frac{2}{5}\right)^{-1} = \frac{5}{2}$ and the identity exponent law states $\frac{5}{2} = \left(\frac{5}{2}\right)^{1}$, it can be concluded that $\left(\frac{2}{5}\right)^{-1} = \left(\frac{5}{2}\right)^{1}$.

2. Continue the pattern.

Exponent	Expanded Form	Equivalent Expression
$\left(\frac{2}{5}\right)^{0}$	1	$\frac{2^0}{5^0}$
$\left(\frac{2}{5}\right)^{-1}$	$\frac{5}{2}$	$\left(\frac{5}{2}\right)^{1}$
$\left(\frac{2}{5}\right)^{-2}$		
$\left(\frac{2}{5}\right)^{-3}$		

$\times \frac{5}{2}$

$\times \frac{5}{2}$

$\times \frac{5}{2}$

3. Complete the statement.

A rational base with a negative exponent, $\left(\frac{a}{b}\right)^{-m}$, can be rewritten using the

- ○ reciprocal
- ○ opposite

of the base with the

- ○ reciprocal
- ○ opposite

exponent.

4. Rewrite each expression as an equivalent exponential expression with positive exponents.

a. $\left(\frac{2}{3}\right)^{-4} =$ ________

b. $\left(\frac{3}{-5}\right)^{-9} =$ ________

c. $\left(-\frac{8}{9}\right)^{-2} =$ ________

d. $\left(\frac{21}{4}\right)^{-5} =$ ________

4.2.3 Guided Practice: Applying Multiple Exponent Laws to Expressions with Rational Bases

1. The table shows equivalent forms of the exponential expression $\left(\frac{4}{5}\right)^4 \cdot \left(\frac{4}{5}\right)^{-3}$.

   a. Complete the table to find the value of $\left(\frac{4}{5}\right)^4 \cdot \left(\frac{4}{5}\right)^{-3}$.

Exponential Expression	Apply Negative Exponent Law	Expanded Form	Value
$\left(\frac{4}{5}\right)^4 \cdot \left(\frac{4}{5}\right)^{-3}$	$\left(\frac{4}{5}\right)^4 \cdot \left(\frac{5}{4}\right)^{3}$	$\frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{5}{4} \cdot \frac{5}{4} \cdot \frac{5}{4}$	

   b. Explain how you used the expanded form of the expression to determine the value.

Multiple exponent laws can be applied to write equivalent forms of exponential expressions and to evaluate them.

2. For each expression, apply one or more exponent laws to rewrite as an equivalent expression with a rational base and positive exponent. Check off the exponent law(s) you applied to generate the expression. Show your work.

Exponential Expression	Equivalent Expression with a Rational Base and Positive Exponent	Exponent Laws Applied
$\left(\frac{4}{5}\right)^4 \cdot \left(\frac{4}{5}\right)^{-3}$		☐ negative exponent ☐ power of a power ☐ power of a product ☐ product of powers ☐ quotient of powers
$\left(\frac{11}{3}\right)^{-5} \div \left(\frac{11}{3}\right)^2$		☐ negative exponent ☐ power of a power ☐ power of a product ☐ product of powers ☐ quotient of powers
$\left[\left(\frac{2}{4}\right)^2\right]^{-5}$		☐ negative exponent ☐ power of a power ☐ power of a product ☐ product of powers ☐ quotient of powers
$\left(\frac{2}{3} \cdot \frac{4}{5}\right)^{-4}$		☐ negative exponent ☐ power of a power ☐ power of a product ☐ product of powers ☐ quotient of powers

4.2.4 Collaboration: Comparing Strategies

1. Edeline and Franco determined the same equivalent expression for $\left(\frac{7}{11}\right)^{-5}$, but they used different strategies.

Their work is shown in the table.

Edeline's Work	Franco's Work
$\left(\frac{7}{11}\right)^{-5} = \frac{7^{-5}}{11^{-5}} = \frac{11^{5}}{7^{5}}$	$\left(\frac{7}{11}\right)^{-5} = \left(\frac{11}{7}\right)^{5} = \frac{11^{5}}{7^{5}}$

a. Discuss both students' strategies with your partner.

b. What steps did Edeline take to find the equivalent expression?

c. What steps did Franco take to find the equivalent expression?

d. Whose method do you prefer?

4.2.5 Your Turn!

1. **Match each expression in Column A to an equivalent expression in Column B by drawing arrows.**

Column A	Column B
$\left(-\frac{7}{8}\right)^{-2}$	$\left(\frac{7}{8}\right)^{2}$
$\left(-\frac{8}{7}\right)^{-2}$	$\left(-\frac{8}{7}\right)^{2}$
$\left(\frac{7}{8}\right)^{-2}$	$\left(\frac{8}{7}\right)^{2}$
$\left(\frac{8}{7}\right)^{-2}$	$\left(-\frac{7}{8}\right)^{2}$

2. **Three expressions are shown. Circle the two expressions that are equivalent.**

$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^{4}$ $\qquad$ $\dfrac{\left(\frac{5}{8}\right)^{-4}}{\left(\frac{5}{8}\right)^{-6}}$ $\qquad$ $\left(\left(\frac{5}{8}\right)^{-2}\right)^{4}$

3. **Generate an equivalent expression using the laws of exponents. Express your answer using positive exponents.**

$$\left(\frac{9}{10} \cdot \frac{5}{6}\right)^{-6}$$

4.2.6 Wrap-Up: Ticket Out the Door

1. **Rewrite each expression as an equivalent exponential expression with positive exponents.**

   a. $\left(\frac{-12}{2}\right)^{-3} =$ _______________

   b. $\left(\frac{1}{5}\right)^{-8} =$ _______________

2. **Generate an equivalent expression using the laws of exponents. Express your answer with positive exponents.**

$$\left(\frac{-5}{4}\right)^{-7} \cdot \left(\frac{-5}{4}\right)^{-3}$$